

Customer Churn Prediction in Telecom Services (Milestone 5)

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DSC-630: Predictive Analysis

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May 27, 2025

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Introduction

Problem Statement

This project aims to develop a predictive model for customer churn in the telecommunications industry. Customer churn, the rate at which customers stop doing business with a company, represents a significant challenge for telecom providers. By accurately predicting which customers are likely to churn, telecom companies can implement targeted retention strategies before customers leave, thereby reducing revenue loss and maintaining market share.

Importance of the Problem

Customer churn directly impacts a telecom company's bottom line. Industry research indicates that acquiring a new customer costs 5-25 times more than retaining an existing one. Furthermore, increasing customer retention rates by just 5% can increase profits by 25-95%. In the highly competitive telecom market, where customers have multiple service options, predicting and preventing churn is essential for sustainable growth and profitability.

Stakeholders

This project would provide valuable insights to:

- Marketing departments developing retention campaigns
- Customer service teams requiring early warning systems
- Product development teams addressing service gaps

- Executive leadership making strategic business decisions
- Finance departments forecasting revenue and customer lifetime value

Data Source

The primary dataset for this analysis comes from the IBM Sample Data Sets repository, specifically the "Telco Customer Churn" dataset. This dataset contains information for approximately 7,000 telecom customers including:

- Customer demographics (gender, age range, partners, dependents)
- Account information (tenure, contract type, payment method)
- Services subscribed (phone, internet, tech support, online security)
- Billing information (monthly charges, total charges)
- Churn status (whether the customer left within the last month)

Data Relevance

This dataset is particularly well-suited for the problem for several reasons:

1. It provides comprehensive customer attributes across multiple dimensions
2. It includes labeled outcomes (churned/not churned), enabling supervised learning
3. It contains both categorical and numerical features, allowing for rich pattern discovery
4. The data represents actual telecom customer behavior patterns
5. The size of the dataset is sufficient for training reliable models while avoiding computational limitations

Proposed Methods

I plan to implement and compare the following classification models:

1. Logistic Regression: As a baseline model to identify key predictors and provide interpretable results. This will establish benchmark performance metrics and feature importance.
2. Random Forest: To capture non-linear relationships and complex interactions between features. The ensemble approach will help reduce overfitting and handle the categorical variables present in the dataset.
3. Gradient Boosting (XGBoost): For potentially higher predictive performance through sequential learning. This advanced technique can capture subtler patterns in the data.
4. Neural Networks: If warranted by initial results, to capture more complex underlying patterns that traditional methods might miss.

The selection of these models provides a progression from simple, interpretable models to more complex, potentially more accurate approaches.

Evaluation Strategy

Given the business context of customer churn prediction, the evaluation will prioritize:

1. Recall (Sensitivity): The proportion of actual churners correctly identified is critical, as missing a potential churner is costlier than falsely flagging a loyal customer.
2. AUC-ROC: To evaluate the model's ability to distinguish between churners and non-churners across various threshold settings.
3. Precision: To ensure retention resources aren't wasted on false positives.
4. F1-Score: For a balanced evaluation of precision and recall.

5. Business Impact Analysis: A custom metric incorporating the estimated cost of retention campaigns versus the revenue saved by preventing churn.

Expected Insights

Through this project, I hope to:

1. Identify the primary drivers of customer churn in telecom services
2. Quantify the impact of different service combinations on retention
3. Understand how contract types and billing methods influence customer loyalty
4. Develop actionable customer segments based on churn propensity
5. Create a framework for ongoing churn prediction that can be updated as new data becomes available

Risk Assessment and Ethics

Potential Risks

This project is prone to the standard risks of actionable data science attempts:

1. Data Quality Issues: Missing values, outliers, or biased sampling could affect model performance
2. Feature Selection Challenges: Determining the most relevant predictors without overfitting
3. Model Drift: Changes in customer behavior patterns over time may require model retraining
4. Implementation Complexity: Integrating the predictive model into existing business processes

Ethical Considerations

1. Privacy Concerns: Customer data must be handled responsibly and in compliance with regulations like GDPR or CCPA
2. Fairness: The model should not discriminate against specific demographic groups
3. Transparency: Predictions should be interpretable to both business users and customers
4. Customer Autonomy: Respect for customer choices rather than manipulative retention tactics

I will address these ethical considerations by:

- Implementing fairness metrics during model evaluation
- Using explainable AI techniques to provide insight into model decisions
- Focusing on creating value for both customers and the company
- Considering differential privacy techniques when working with sensitive personal data

Contingency Plan

If the initial approach does not yield satisfactory results, I will:

1. Explore additional data sources that might provide better predictors of churn
2. Implement more advanced feature engineering techniques
3. Consider ensemble methods combining multiple model types
4. Shift focus to specific customer segments where prediction accuracy is higher
5. Reframe the problem as a customer lifetime value prediction if binary churn prediction proves challenging

Preliminary Analysis

Key Insights and Business Recommendations

My EDA of the Telco customer churn dataset revealed several important insights that can guide business strategies for customer retention:

1. **Contract Type is Crucial:** Month-to-month contracts have significantly higher churn rates (~43%) compared to one-year (~11%) and two-year contracts (~3%). This is one of the strongest predictors of churn in the dataset.
2. **Early Tenure Risk:** Customers with shorter tenure (0-12 months) show much higher churn rates, particularly those on month-to-month contracts. The first year appears to be a critical period for customer retention.
3. **Fiber Optic Service Issues:** Customers with fiber optic internet service churn at more than twice the rate of DSL customers, suggesting potential issues with this service offering.
4. **Additional Services as Retention Tools:** Customers with supplementary services like tech support, online security, and online backup show significantly lower churn rates.
5. **Payment Method Signals:** Electronic check payment is associated with much higher churn rates compared to other payment methods.
6. **Senior Citizen Vulnerability:** Senior citizens have higher churn rates than non-seniors, possibly indicating challenges with service usability or satisfaction.
7. **Price Sensitivity:** Higher monthly charges correlate with increased churn, particularly among newer customers.

Answering Research Questions

Based on the EDA conducted, I'll address the following key research questions:

1. Will I be able to answer the questions I want to answer with the data I have?

The dataset appears to be well-suited for predicting customer churn and identifying key factors that contribute to churn. Key findings that support this conclusion include:

- The dataset has a good mix of demographic, account, and service-related features
- Clear patterns are visible in the relationship between several features and churn status
- The preliminary feature importance analysis shows promising predictive power for several variables
- There is minimal missing data, making the dataset relatively clean for modeling

2. What visualizations are especially useful for explaining my data?

The most insightful visualizations for this project include:

- Contract type vs. churn status - Shows a strong relationship that can be easily communicated to stakeholders
- Tenure distribution by churn status - Clearly illustrates how newer customers are more likely to churn
- Service subscription patterns vs. churn - Highlights which services are associated with higher retention
- Scatter plot of tenure vs. monthly charges - Reveals interaction effects between these important variables

- Feature importance chart - Provides a clear ranking of predictive factors for non-technical audiences

3. Do I need to adjust the data and/or driving questions?

Based on the EDA, some adjustments that would be beneficial include:

- Create derived features that combine related services to better capture customer profiles
- Transform tenure into categorical buckets for easier interpretation
- Consider the ratio of monthly charges to services as a potential predictor of value perception
- The "SeniorCitizen" variable should be converted to categorical format for consistency

4. Do I need to adjust my model/evaluation choices?

The EDA suggests the following adjustments to modeling approach:

- The moderate class imbalance in the churn variable (~27% churn rate) confirms the need to prioritize recall and potentially use techniques like SMOTE or class weighting
- The strong feature importance of contract type and tenure suggests that simpler models might perform well
- Including both total charges and tenure might introduce multicollinearity, so careful feature selection or regularization will be important
- Decision tree-based models seem appropriate given the mix of categorical and numerical features and potential non-linear relationships

5. Are my original expectations still reasonable?

The original expectations for the project are still reasonable based on the EDA. The data does contain strong signals for predicting churn, and the identified patterns align with industry knowledge about customer retention. The proposed modeling approaches (Logistic Regression, Random Forest, Gradient Boosting, and potentially Neural Networks) are appropriate given the dataset characteristics.

Next Steps for Modeling

Based on my exploratory data analysis, I can proceed to predictive modeling with the following approach:

1. **Feature Selection:** Prioritize the strong predictors identified in our analysis:

- Contract type
- Tenure
- Internet service type
- Additional services (tech support, online security)
- Payment method
- Monthly charges

2. **Feature Engineering:**

- Create customer segments based on tenure and contract type
- Develop a service bundle indicator
- Consider interaction terms between tenure and contract type

- Transform right-skewed variables like TotalCharges
- Calculate price per service metrics
- Transform categorical variables appropriately for modeling

3. **Modeling Strategy:**

- Build classification models (logistic regression, random forest, gradient boosting)
- Use stratified cross-validation due to the class imbalance (26.5% churn rate)
- Optimize for business-relevant metrics such as recall or precision, depending on the cost of false negatives vs. false positives
- Address class imbalance through appropriate techniques
- Develop a strong baseline model using logistic regression

4. **Model Interpretation and Evaluation:**

- Focus on understanding feature importance
- Develop actionable insights for different customer segments
- Create churn probability scores for targeting retention efforts
- Prioritize recall while maintaining acceptable precision
- Calculate the business impact of different threshold choices

5. **Business Application:**

- Begin formulating targeted strategies for high-risk customer groups
- Identify the most cost-effective interventions based on feature importance

- Prepare to translate model insights into actionable recommendations

Methods and Results

Data Preparation

The final modelling phase used the publicly available Telco Customer Churn CSV from IBM (IBM, 2024). After the exploratory work described above, the data was prepared as follows:

1. Removed the non-informative customerID field.
2. Converted TotalCharges to numeric, coercing errors and imputing blank entries with 0.
3. Binary-encoded the target (Churn: Yes $\rightarrow$ 1, No $\rightarrow$ 0).
4. Separated the four numerical predictors (SeniorCitizen, tenure, MonthlyCharges, TotalCharges) from the 15 categorical predictors listed in the EDA.
5. Built a preprocessing pipeline with:
 - a. StandardScaler for numeric columns
 - b. OneHotEncoder(handle_unknown="ignore") for categoricals.

Modelling Strategy

Recent churn studies report strong baselines from regularized Logistic Regression (Chang et al., 2024; Chajia & Nfaoui, 2024). Consistent with this literature, and to retain full interpretability, I trained a class-weighted Logistic Regression model inside the preprocessing pipeline. A stratified 75 / 25 split was used ($n = 5,282$ train, 1,761 test), and model hyper-parameters were tuned with 5-fold cross-validation on the training set.

Performance

Metric	CV Mean $\pm$ SD	Hold-out Test
ROC-AUC:	0.844 $\pm$ 0.013	0.846
Accuracy:	-	0.75
Recall (churn):	-	0.80
Precision (churn):	-	0.52

The area-under-curve of ≈ 0.85 indicates excellent rank-ordering of customers by churn risk. Precision is lower because the minority class is only $\sim 27\%$ of the sample, but recall of 0.80 means eight out of ten churners are captured—an attractive trade-off for retention campaigns where the cost of a false positive is modest.

Interpretation

Model coefficients reinforce the patterns discovered during EDA:

- Long tenure ($\beta = -1.15$) and two-year contracts strongly reduce churn odds.
- Month-to-month contracts and fiber-optic internet increase churn propensity.
- Higher monthly charges are associated with elevated risk, while add-on security / support services are retention levers.

These signals align with prior work showing contract structure and perceived value as primary churn drivers (Chang et al., 2024).

Conclusion

Using the IBM Telco data, I built an end-to-end, interpretable Logistic Regression pipeline that delivers ROC-AUC ≈ 0.85 on unseen data. The analysis confirms the following points:

- Short-tenure, month-to-month subscribers paying higher fees are at the greatest risk of attrition
- Extended contracts and value-added services materially improve retention.

Thus, my business recommendations based on the analysis completed in this project are as follows:

1. Proactively target new (< 12 month) month-to-month customers with loyalty incentives or contract upgrades.
2. Bundle fiber-optic access with discounts or service credits to raise perceived value.
3. Continue promoting support / security add-ons that correlate with lower churn.

Further work could compare tree-based ensembles and calibrate a cost matrix that monetizes false-positive vs. false-negative outcomes.

References

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